Gage DeZoort

Postdoctoral Research Associate

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Education

Princeton University *Princeton, NJ*

PhD in Physics *2023*

- Center for Statistics and Machine Learning Certificate
- Advisor: Daniel Marlow

University of Virginia *Charlottesville, VA*

BS in Physics and Engineering Science *2018*

- With Highest Distinction
- Jefferson Scholar and Rodman Scholar

Publications and Proceedings

G. DeZoort and B. Hanin (2026). Hyperparameter Transfer in Graph Neural Networks. Accepted for publication in the proceedings of the Learning on Graphs 2026 conference. Preprint: arXiv:2607.05017.

The CMS Collaboration (2025). Search for a heavy pseudoscalar Higgs boson decaying to a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons in proton-proton collisions at $\sqrt{s}$ = 13 TeV. Journal of High Energy Physics. `https://doi.org/10.1007/JHEP10(2025)074`.

G. DeZoort and Boris Hanin (2025). Principles for Initialization and Architecture Selection in Graph Neural Networks with ReLU Activations. SIAM Journal of Mathematics of Data Science. `https://doi.org/10.1137/23M1600621`.

K. Lieret, **G. DeZoort**, et al. (2023). High Pileup Particle Tracking with Object Condensation. Proceedings for the 8th International Connecting the Dots Workshop. arXiv:2312.03823.

K. Lieret and **G. DeZoort** (2023). An Object Condensation Pipeline for Charged Particle Tracking at the High Luminosity LHC. Proceedings for the 26th International Conference on Computing in High Energy and Nuclear Physics (CHEP23). `https://doi.org/10.1051/epjconf/202429509004`.

G. DeZoort et al. (2023). Graph Neural Networks at the Large Hadron Collider. Nat Rev Phys 5, 281–303 (2023). `https://doi.org/10.1038/s42254-023-00569-0`.

S. Thais et al. (2022). Graph Neural Networks in Particle Physics: Implementations, Innovations, and Challenges. Contribution to the 2022 Snowmass Summer Study. arXiv:2203.12852.

C. Wang, X. Ju, et al. (2022). Reconstruction of Large Radius Tracks with the Exa.TrkX Pipeline. Proceedings of 20th International Workshop on Advanced Computing and Analysis Techniques in Physics Research. arXiv:2203.08800.

A. Lazar, X. Ju, D. Murnane et al. (2022). Accelerating the Inference of the Exa.TrkX Pipeline. Proceedings of 20th International Workshop on Advanced Computing and Analysis Techniques in Physics Research. arXiv:2202.06929.

A. Elabd, V. Razavimaleki et al. (2022). Graph Neural Networks for Charged Particle Tracking on FPGAs. Frontiers in Big Data Science, Sec. Big Data and AI in High Energy Physics. doi:10.3389/fdata.2022.828666.

X. Ju, D. Murnane, et al. (2021). Performance of a geometric deep learning pipeline for HL-LHC particle tracking. European Physics Journal C 81(876). doi:10.1140/epjc/s10052-021-09675-8.

G. DeZoort et al. (2021). Charged Particle Tracking via Edge-Classifying Interaction Networks. Computing and Software for Big Science 5(26). doi:10.1007/s41781-021-00073-z.

A. Heintz, V. Razavimaleki et al. (2020). Accelerated Charged Particle Tracking with Graph Neural Networks on FPGAs. Third Workshop on Machine Learning and the Physical Sciences (NeurIPS 2020). arXiv:2012.01563.

The CMS Collaboration (2018). Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}$ = 13 TeV. Journal of High Energy Physics. doi:10.1007/JHEP11(2018)042.

In Review / Preparation

Y. Dong, **G. DeZoort**, and M. Li (2026). Learning Rate Transfer for Reinforcement Learning Post-Training. Submitted to the Fifteenth International Conference on Learning Representations.

G. DeZoort, A. Frankenthal, and B. Hanin (2026). Identifying Jets with Coupled Particle-Edge Networks. In preparation.

Awards, Fellowships, & Grants

2025-2026	**School of Physical Sciences Visiting Fellowship**	*UC Irvine*
2025	**Breakthrough Prize in Fundamental Physics**	*Awarded to the CMS Collaboration*
2022	**Kusaka Memorial Prize in Physics**	*Princeton Physics*
	Departmental Equity, Diversity, and Inclusion Award	*Princeton Physics*
2021	**Departmental Equity, Diversity, and Inclusion Award**	*Princeton Physics*
2020	**Departmental Teaching Award**	*Princeton Physics*
2017	**Mitchell Research Scholarship**	*University of Virginia*
2016	**Mitchell Research Scholarship**	*University of Virginia*
2014	**Rodman Scholarship**	*University of Virginia*
	Jefferson Scholarship	*University of Virginia*

Presentations

G. DeZoort (Nov. 2026). *Hyperparameter Transfer in Graph Neural Networks*. Poster: Learning on Graphs 2026. Boston, Massachusetts.

G. DeZoort (Nov. 2026). *Initialization and Architecture Selection for GNNs*. Invited Talk: SIAM Conference on Mathematics of Data Science. Salt Lake City, Utah.

G. DeZoort (2026). *Hyperparameter Transfer: A Recipe for Efficient and Robust Scaling*. Plenary Talk: HAMLET-PHYSICS 2026. Copenhagen, Denmark.

G. DeZoort and B. Hanin (2025). *Hyperparameter Transfer for Graph Transformers*. Talk: 23rd International Workshop on Advanced Computing and Analysis Techniques in Physics Research. Hamburg, Germany.

G. DeZoort et al. (2024). *Search for a heavy pseudoscalar Higgs boson decaying to Zh with $ll\tau\tau$ final states*. Approval talk for the HIG-22-004 analysis.

G. DeZoort and Boris Hanin (2024). *Principled Graph Neural Networks*. Poster: USCMS Annual Meeting, Princeton, NJ.

G. DeZoort et al. (2022). *Particle Tracking with Graph Neural Networks*. Talk: Learning to Discover Workshop, Orsay, France.

G. DeZoort et al. (2022). *Track Condensation Networks*. Talk: Connecting the Dots Workshop, Princeton, NJ.

G. DeZoort et al. (2022). *The Princeton Physics Ambassadors Program.* Talk: 2022 April American Physical Society Meeting, New York City, NY.

G. DeZoort et al. (2021). *Charged Particle Tracking via Edge-Classifying Interaction Networks*. Poster: Princeton Physics Dicke Fellows Symposium, Princeton, NJ.

G. DeZoort et al. (2021). *Charged Particle Tracking via Edge-Classifying Interaction Networks*. Talk: Oxford School of ML, held remotely.

G. DeZoort et al. (2021). *Charged Particle Tracking via Edge-Classifying Interaction Networks*. Poster: London School of Geometry and ML, held remotely.

G. DeZoort et al. (2021). *Charged Particle Tracking via Edge-Classifying Interaction Networks*. Plenary Talk: Virtual Conference on Computing in High Energy and Nuclear Physics, held remotely.

Savannah Thais and **G. DeZoort** (2020). *Instance Segmentation GNNs for One-Shot Conformal Tracking*. Poster: NeurIPS Machine Learning and the Physical Sciences workshop, held remotely.

G. DeZoort et al. (2020). *Particle Tracking with Graph Neural Networks Accelerated on FPGAs*. Contributed Talk: Accelerated Artificial Intelligence for Big-Data Experiments conference, held remotely.

G. DeZoort and Dan Marlow (2019). *Desktop Cosmic Particle Detector*. Poster: Princeton Research Day, Princeton, NJ.

G. DeZoort et al. (2017). *The CMS Electron and Photon Trigger for the LHC Run 2*. Poster: American Physical Society April Meeting, Washington, DC.

G. DeZoort et al. (2016). *Anomalous Signal Reduction in the CMS Detector's ECAL Trigger*. Poster presentation: Society of Physics Students Quadrennial Physics Congress, San Francisco, CA. Outstanding Poster Presentation award in general physics category.

G. DeZoort et al. (2016). *Anomalous Signal Reduction in the CMS Detector's ECAL Trigger*. Contributed Talk: Southeastern Section of the APS Annual Conference, Charlottesville, VA.

Teaching

Fall 2024	**PHY 101: Introductory Physics I** Instructor	*Princeton, NJ*
Fall 2023	**PHY 103: General Physics I** Instructor	*Princeton, NJ*
Fall 2022	**PHY 103: General Physics I** Instructor	*Princeton, NJ*
Spring 2020	**PHY 529: High Energy Physics** Teaching Assistant	*Princeton, NJ*
Fall 2019	**PHY 115: Physics for Future Leaders** Teaching Assistant	*Princeton, NJ*
Spring 2018	**PHY 2660: Introduction to Scientific Computing** Teaching Assistant	*Charlottesville, VA*
Fall 2017	**PHY 1010: The Physical Universe** Teaching Assistant	*Charlottesville, VA*
Fall 2017	**PHY 1910: Intro to Physics Research** Course Organizer	*Charlottesville, VA*
Spring 2017	**PHY 2660: Intro to Scientific Computing** Teaching Assistant	*Charlottesville, VA*

Tutorials and Workshops

Oct. 2025	**Crash Course in Practical ML** Instructor `https://github.com/GageDeZoort/intro_ml_uci/`	*Irvine, CA*
January 2025	**Introduction to Machine Learning** Instructor `https://github.com/PrincetonUniversity/intro_machine_learning`	*Princeton, NJ*
March 2024	**The Neural Network Zoo** Instructor `https://github.com/GageDeZoort/neural-network-zoo`	*Princeton, NJ*
January 2024	**Introduction to Machine Learning** Instructor `https://github.com/PrincetonUniversity/intro_machine_learning`	*Princeton, NJ*
Spring 2023	**Graph Neural Networks for Your Research** Workshop Instructor `https://github.com/GageDeZoort/prc_gnn_tutorial`	*Princeton, NJ*
January 2023	**Introduction to Machine Learning** Instructor `https://github.com/PrincetonUniversity/intro_machine_learning`	*Princeton, NJ*
July 2022	**Introduction to Graph Neural Networks with PyG** Workshop Instructor `https://indico.cern.ch/event/1151329/`	*Princeton, NJ*

Mentoring

2025	**Alice Hu** Princeton University OURSIP Summer Research Scholar: *Data Augmentation for Graph Transformers*	*Princeton, NJ*
2024-2025	**Joah Macosko** Princeton University Senior Thesis: *Machine Learning for Extracting Track Physics*	*Princeton, NJ*
Spring 2024	**Joah Macosko** Princeton University Junior Project: *Using Machine Learning Techniques to Measure Track Physics in High-Pileup Collision Events*	*Princeton, NJ*
2021-2022	**Mufaro Chitoto** Princeton University Senior Thesis: *Charged Particle Tracking with Interaction Networks*	*Princeton, NJ*

Outreach & Professional Development

Service and Outreach

2025-2026	**Machine Learning for Jet Physics Reading Group,** Organizer	*Irvine, CA*
2023-2024	**Machine Learning in Physics Seminar Series,** Organizer	*Princeton, NJ*
2023-2024	**Princeton Physics EDI,** Postdoctoral Matters Committee Member	*Princeton, NJ*
2023	**Prospective Physics Ph.D. Preview (P4) Workshop,** Organizer	*Princeton, NJ*
2022-2023	**Princeton Physics Graduate Committee,** 5th-Year Class Representative	*Princeton, NJ*
2022	**Prospective Physics Ph.D. Preview (P4) Workshop,** Lead Organizer	*Princeton, NJ*
2021-2022	**Princeton Physics Graduate Committee,** 4th-Year Class Representative	*Princeton, NJ*
2021	**Prospective Physics Ph.D. Preview (P4) Workshop,** Lead Organizer	*Princeton, NJ*
2020-2023	**Princeton Physics Ambassadors (EDI Initiative),** Founder and President	*Princeton, NJ*
2020-2022	**Princeton Physics EDI,** Graduate Matters Committee Member	*Princeton, NJ*
2020-2022	**McGraw Center for Teaching and Learning,** Graduate Teaching Fellow	*Princeton, NJ*
2019-2020	**Graduate Student Government,** Physics Department Representative	*Princeton, NJ*

Professional Development

2022	**CERN-Fermilab Hadron Collider Physics Summer School**	*Virtual*
2021	**Oxford Machine Learning Summer School**	*Virtual*
2021	**Machine Learning in High Energy Physics Summer School**	*Virtual*
2021	**London Geometry and Machine Learning Summer School**	*Virtual*
2019	**CMS Data Analysis School**	*Fermilab*
2018	**Computational and Data Science in HEP**	*Princeton, NJ*
2017	**CMS Data Analysis School**	*Fermilab*

Professional Memberships

Sigma Pi Sigma Physics Honor Society
University of Virginia Raven Society